

A Constant Vector Beats Your Digital Twin: On the Evaluation of Cluster-Level LLM Personality Simulation

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Abstract

Large language models are increasingly prompted to act as “digital twins” of human personality groups, and such systems are routinely evaluated by how closely the model’s questionnaire answers match those of real respondents. We show that this evaluation is broken. For the answer-similarity metric used throughout this literature, we prove that a simulator which reproduces the target population’s response distribution *exactly* is strictly dominated by a constant vector, and that the gap is governed by the difference between the Gini mean difference and the mean absolute deviation about the median. On 410,168 IPIP-NEO-120 respondents this structural penalty is 7.75 points. Empirically, across five LLMs and four personality clusters, a per-item cluster mean scores 0.796 while evolved personas average 0.739, losing in 20 of 20 cells; a *real human drawn from the target cluster* scores 0.721 and also loses to the constant in 20 of 20 cells. We further find that the released benchmark scores raw model self-reports against reverse-recoded human answers on 55 of 120 items; we correct this and report every result under both orientations, the replacement suite included. A metric that ranks a constant above a genuine group member cannot be measuring personality simulation. We propose a replacement suite — a classifier two-sample test, per-item distributional fidelity, and a variance ratio — each reported against an explicit human ceiling and constant floor, and show it inverts the ranking correctly. Under it, current personas are near-perfectly separable from humans (C2ST-AUC 0.994 vs. a human ceiling of 0.564) and reach only 0.43 of human response *spread* as a standard-deviation ratio. Correct persona assignment improves CRPS for all eleven models, yet all lose to a cluster empirical prior fitted on other respondents, and an exact decomposition shows the excess loss of the model’s mixture dominates the shortfall. We release code and per-cell results.

1 Introduction

Simulating human respondents with large language models has become a common instrument in computational social science and in LLM-based user research. A widely used recipe conditions a model on a natural-language personality description and validates it by administering a psychometric inventory: the model answers the same Likert items a human answered, and the system is scored by the agreement between the two answer vectors.

This paper argues that the scoring step invalidates the conclusion. Our starting point is a concrete artefact. In a cluster-level digital-twin system built on IPIP-NEO-120 (Johnson, 2014), evolutionary prompt optimisation raises answer-level agreement while simultaneously *degrading* the recovered Big Five profile — mean absolute error, trait similarity and facet similarity all move the wrong way. That pattern is usually reported as an open puzzle. We show it is a predictable consequence of the metric.

Our contributions are:

1. **A proposition (§3):** under answer-similarity, a simulator that faithfully samples the target population’s response distribution is *dominated* by the best constant predictor — strictly so on every IPIP-NEO item we measure. The penalty is the excess of the Gini mean difference over the mean absolute deviation about the median, divided by the scale range.
2. **An empirical audit (§4):** across five LLMs $\times$ four clusters, on the original system’s own train/test splits, the matched cluster-mean baseline beats every model in every cell, and a global mean that ignores the clusters entirely beats it in 19 of 20.
3. **A diagnosis of the human ceiling (§7):** a real respondent from the target cluster — the ideal

a cluster-level twin can aspire to — also loses to the constant in every cell.

4. **A replacement suite** (§8): C2ST-AUC, distributional fidelity and variance ratio, each anchored by a human ceiling and a constant floor. It ranks humans above constants, as any valid measure must.
5. **A cluster-validity audit** (§12): the $k=4$ partition these systems condition on is not supported by any internal criterion.

2 Setting

Task. A population of respondents is partitioned into K personality clusters by k -means over the 30 IPIP-NEO facet scores. For each cluster a natural-language description of the centroid is used to condition an LLM, which then answers the 120 IPIP-NEO items on a 1–5 Likert scale. Prompt wording is optimised by an evolutionary search whose fitness is answer agreement with held-out human members of the cluster.

The metric. A simulated answer vector x is scored against a human answer vector y by

$$S(x, y) = \frac{1}{J} \sum_{j=1}^J \left(1 - \frac{|x_j - y_j|}{4}\right), \quad (1)$$

with $J=120$ and the divisor 4 the scale range. Cell scores average S over the held-out respondents. Equation (1), or a monotone transform of it, is the objective the search optimises and the number reported.

3 A constant is optimal under S

Fix a cluster and an item j , and let P_j be the cluster’s response distribution over $\{1, \dots, 5\}$. Write $Y \sim P_j$ for a randomly drawn human respondent.

Proposition 1. *Let m_j be a median of P_j , let $\text{MAD}_j = \mathbb{E}_Y |m_j - Y|$ and let $\text{GMD}_j = \mathbb{E}_{X,Y} |X - Y|$ for $X \sim P_j$ independent of Y (the Gini mean difference). Then*

- (a) *the best constant predictor attains expected score $1 - \text{MAD}_j/4$;*
- (b) *a faithful simulator, i.e. one whose output is distributed as P_j , attains expected score $1 - \text{GMD}_j/4$;*
- (c) *$\text{GMD}_j \geq \text{MAD}_j$, with equality iff X lies almost surely in the median set of P_j .*

Hence the constant is at least as good as the faithful simulator, and strictly better unless P_j places all its mass on medians.

Remark 1 (Where this applies). *Proposition 1 is unconditional; the audited systems condition on something. Let C be everything the simulator observes, its draw independent of the response given C . Under (1) an optimal simulator emits a conditional median $m_j(C)$, so one constant stays optimal exactly when the conditional median sets share a value almost surely, item by item. Cluster-only conditioning thus admits a cluster-specific constant optimum; individually informative conditioning can make different deterministic answers optimal, so the constant need not win. A deterministic conditional median is always optimal and a randomised predictor ties exactly when its support lies in that median set, so absolute error never uniquely rewards faithful conditional sampling; richer conditioning moves which prediction is rewarded.*

Proof. (a) and (b) are immediate from (1). For (c),

$$\begin{aligned} \text{GMD}_j &= \mathbb{E}_X [\mathbb{E}_Y |X - Y|] \\ &\geq \mathbb{E}_X [\min_c \mathbb{E}_Y |c - Y|] \\ &= \mathbb{E}_Y |m_j - Y| = \text{MAD}_j, \end{aligned}$$

since the median minimises expected absolute deviation. Equality requires X to attain that minimum almost surely, i.e. to lie in the median set of P_j . This is weaker than degeneracy: a balanced two-point law $P(Y=1) = P(Y=5) = \frac{1}{2}$ has $\text{GMD} = \text{MAD} = 2$, so the bound is tight there despite large variance. On the IPIP items it is strict throughout (Table 2). $\square$

The consequence is not a technicality. Under S , the ideal behaviour we actually want — reproducing the population’s response distribution — is penalised relative to emitting one fixed vector. Optimising S therefore pushes a simulator toward degenerate central-tendency responding, destroying exactly the answer variance that carries a personality profile. This is the mechanism behind the “answer similarity up, profile fidelity down” pattern reported as a puzzle in prior work.

3.1 The metric is CRPS with its dispersion term deleted

Proposition 1 is not an accident of Likert data. The continuous ranked probability score for a predictive distribution F and outcome y is

$$\text{CRPS}(F, y) = \mathbb{E}_F |X - y| - \frac{1}{2} \mathbb{E}_F |X - X'|,$$

λ	Model	Constant	Human
0 (audited)	0.726	0.796	0.720
0.25	0.749	0.796	0.789
0.5 (CRPS, proper)	0.772	0.796	0.859
0.75	0.795	0.796	0.929
1.0	0.819	0.796	0.998

Table 1: The S_λ family of (2), averaged over 20 cells, all values on the corrected orientation. A constant wins at the audited $\lambda=0$; at the proper $\lambda=1/2$ a real cluster member wins, as it must. All-pairs scores, slightly below the paired ones of §4; the ordering is the same under either.

and it is *strictly proper*: uniquely optimised by $F = P$. Equation (1) is its first term alone. Deleting the second is exactly what makes a constant optimal.

This yields the natural repair. Define

$$S_\lambda = 1 - [\mathbb{E}|X - y| - \lambda \mathbb{E}|X - X'|]/4. \quad (2)$$

$\lambda=0$ is the audited metric — improper, constant-optimal. $\lambda=1/2$ is CRPS — strictly proper, truth-optimal. $\lambda>1/2$ is improper in the opposite direction. Crucially $\lambda=1/2$ is **not** a tunable hyperparameter but the unique proper choice, so the repair cannot be dismissed as a metric tuned to produce a preferred conclusion.

Sweeping λ over the same 20 cells (Table 1) shows the repair works and what it costs the simulator. At $\lambda=1/2$ the human ceiling (0.859) rises decisively above the constant floor (0.796): the ordering that §7 shows is inverted under the audited metric is *correct* under the proper one. Humans overtake the constant at $\lambda^*=0.30$, comfortably below the proper coefficient. The model needs $\lambda^*=0.75$ (median grid crossover; 0.757 on the aggregate curve) — a rule that over-rewards dispersion — and beats the constant in only 2 of 20 cells at the proper $\lambda=1/2$, though it does overtake in most cells once λ reaches 1, where the reward for dispersion is twice proper. So the simulator’s failure is not an artefact of a bad metric: it survives the correct one.

Table 2 evaluates the bound on the real data: the structural penalty against a faithful simulator averages **7.75 points**, and $\text{GMD}_j \geq \text{MAD}_j$ holds on 120/120 items in all four clusters.

4 Empirical audit

Setup. We re-score a published cluster-level digital-twin system on its own artefacts: the same 410,168-responder IPIP-NEO-120 corpus, the

Clu.	n	MAD	GMD	S_{const}	S_{faith}
0	107,174	0.804	1.107	0.7990	0.7232
1	108,880	0.723	1.008	0.8192	0.7480
2	108,867	0.809	1.128	0.7978	0.7181
3	85,247	0.871	1.204	0.7822	0.6990
Mean		0.802	1.112	0.7996	0.7221

Table 2: Proposition 1 on 410,168 IPIP-NEO-120 respondents. $S(\text{const})$ is the best constant predictor; $S(\text{faithful})$ is a simulator that reproduces the cluster response distribution exactly. The faithful simulator is worse in every cluster.

same $k=4$ clusters, and the exact train/test respondent IDs it recorded, for five LLMs (GigaChat3-10B, GPT-4.1-mini, GPT-4.1-nano, Grok-4.1-fast, Qwen3-235B) $\times$ four clusters. We add no LLM calls: model answers are read from the released run artefacts. Baselines are fitted on the recorded train IDs and scored on the recorded test IDs.

Baselines. *Cluster mean*: the per-item rounded mean of the cluster’s training respondents, one constant vector. *Cluster majority*: the per-item mode. *Global mean*: the per-item grand mean over all respondents, ignoring clusters. *Empirical sample*: sample each item independently from the cluster’s training response distribution. *Real human*: copy the answer vector of a randomly chosen training respondent from the same cluster.

Result. Table 3 reports S . The evolutionary persona averages 0.660 as released and 0.739 after the orientation correction of §6; the constant cluster mean averages 0.796. The LLM loses in 20 of 20 cells under either orientation, by 13.6 and 5.8 points respectively. What survives correction is *matched-cell* dominance: the model loses to its own cluster’s constant in every cell, while its best cell (0.807) now exceeds the weakest constant elsewhere (0.779), so the comparison only holds within a cluster. A per-item global mean — which never looks at the cluster at all — scores 0.774, still above the persona’s corrected 0.739.

5 Does conditioning on the right cluster matter?

No persona study we know of reports the wrong-persona control, so no reported persona effect is currently attributable to the persona. We score every simulator against every cluster’s humans — 80 cells — and report diagonal dominance descriptively.

Model	Cluster				Mean
	0	1	2	3	
GPT-4.1-mini	0.657	0.638	0.710	0.652	0.664
GPT-4.1-nano	0.673	0.678	0.697	0.670	0.679
GigaChat3-10B	0.711	0.691	0.737	0.672	0.703
Grok-4.1-fast	0.598	0.583	0.659	0.614	0.613
Qwen3-235B	0.634	0.615	0.677	0.642	0.642
<i>Cluster mean (constant)</i>	0.794	0.827	0.785	0.779	0.796
<i>Real human, same cluster</i>	0.720	0.747	0.715	0.704	0.721

Table 3: Answer similarity S on the original system’s own test IDs, in the *original* orientation; correction adds 0.079 on average and preserves all 20 matched constant wins, though it does reorder the models. The constant (bold) beats every LLM in every cell; a real human scores below it.

Metric	Matched	Mismatched	Δ
S	0.728	0.663	+0.065
DF	0.782	0.705	+0.077
C2ST	0.996	0.985	−0.010
VR	0.431	0.436	−0.005

Matched comparisons improve S and DF; they show no improvement in C2ST or VR. We report these gaps descriptively and attach no p -value: a permutation that shuffles the 80 cell labels freely destroys the matched structure of the five 4×4 matrices, and the models share clusters and personas, so its null is not the right one. Rescoring fixed outputs against different human populations also does not isolate a causal effect of changing the prompt. This is the behavioural counterpart of §8 — the persona moves which items get endorsed, and not who is endorsing them — and it makes the wrong-persona control as necessary a baseline as the constant floor.

6 An orientation defect, and what survives it

Before comparing anything, the two sides must be on the same scale. They are not.

The released corpus stores human answers *reverse-recoded* so that a higher number always means more of the trait, while the simulator is shown the literal item text — “Cheat to get ahead” — and asked how accurately it describes it, i.e. it answers *raw*. On the 55 negatively-keyed items of the IPIP-NEO-120 the two scales run in opposite directions, and every published model-vs-human comparison is computed across that flip.

Two signatures make this unambiguous. Negatively-keyed items average 3.373 against 3.425 for positively-keyed ones — level, whereas raw self-reports on undesirable content sit far lower. And “Cheat to get ahead” carries a human

mean of 4.15 out of 5.

The effect is large and it runs *against* the simulator. Correcting the orientation raises the correlation between the model’s and the humans’ per-item mean profiles from +0.09 — indistinguishable from no relationship at all — to +0.63, and raises S from 0.660 to 0.739.

What matters is which conclusions survive. Three do, and the invariances are stated before the numbers rather than after:

- **Proposition 1 is invariant.** MAD and GMD are computed on the human distribution alone, and both are reflection-invariant.
- **The variance ratio is exactly invariant,** since $\text{sd}(6 - x) = \text{sd}(x)$. The under-dispersion finding of §8 is untouched.
- **The constant still wins every cell.** Reorientation closes the gap from 13.6 to 5.8 points, but the constant cluster mean beats the corrected model in 20 of 20 cells.

Correcting the defect in fact *sharpens* the paper’s claim. The corrected model scores 0.739, *above* the 0.721 of a real member of the target cluster — while a classifier separates that same model from real members at AUC 0.994 (§8). A metric that ranks a near-perfectly detectable simulator above the humans it fails to imitate is not measuring human-likeness under any reading.

We recommend an orientation check as a standing first step for benchmarks of this kind: correlate the simulator’s per-item mean profile against the population’s, and treat a near-zero correlation as a scale defect until shown otherwise.

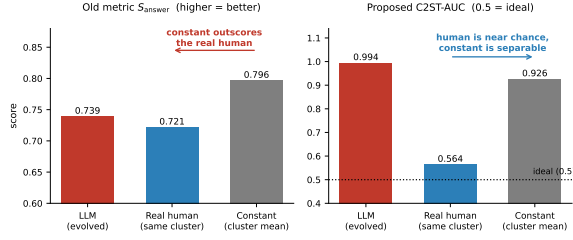


Figure 1: Left: under the old metric the constant outcores a real human. Right: under the proposed C2ST-AUC the human sits near chance (0.5, ideal) and the constant is near-perfectly separable. Only the right panel orders the three correctly.

7 The human ceiling sits below the constant floor

The decisive check is what a *real* member of the target cluster scores. A cluster-level digital twin cannot do better than being an actual member of the cluster; that is the definition of success.

A randomly drawn same-cluster respondent scores 0.721 — and is beaten by the constant cluster mean in 20 of 20 cells (Table 3, Figure 1, left). This is exactly Proposition 1 realised: a real human *is* a draw from P_j , so their expected score is $1 - \text{GMD}/4$, which the constant’s $1 - \text{MAD}/4$ must exceed. The predicted values (0.7996 constant, 0.7221 faithful) match the measured ones (0.796, 0.721) to within 0.4 points.

A measure that ranks a fixed vector above a genuine group member is not measuring group membership. Any ranking it induces over simulators — including the ranking prior work uses to select prompts — is therefore uninterpretable as evidence about personality fidelity.

8 A replacement suite

A usable metric must (i) be maximised by distributional fidelity rather than central tendency, and (ii) be reported against an explicit *human ceiling* and *constant floor*, so a number can be read as progress toward a real target rather than in the abstract. We propose three, computed from a set of simulated respondents rather than from paired individuals.

C2ST-AUC (primary). A classifier two-sample test: train a classifier to separate n simulated answer vectors from n held-out human vectors of the same cluster, and report cross-validated AUC. 0.5 means indistinguishable — the ideal twin. A constant is trivially separable. This directly operationalises the twin claim, and unlike S it cannot be

gamed by collapsing onto the mean.

Distributional fidelity. $1 - \frac{1}{J} \sum_j W_1(\hat{P}_j^{\text{model}}, \hat{P}_j^{\text{human}})/4$, with W_1 the per-item Wasserstein-1 distance. Rewards matching the full response distribution.

Variance ratio. $\frac{1}{J} \sum_j \text{sd}(\text{model}_j)/\text{sd}(\text{human}_j)$; 1.0 is ideal, a constant gives 0. A diagnostic, not an optimisation target.

Results. Table 4 applies the suite to the same 20 cells. All three order human > constant, as required; the old metric does the reverse. Two substantive findings follow. First, current personas are *near-perfectly separable* from real humans (C2ST-AUC 0.994 against a human ceiling of 0.564) — and are in fact *more* distinguishable than a constant vector (0.926), because they miss both the location and the spread of the human distribution. Second, they are systematically under-dispersed: VR = 0.43, a ratio of standard deviations, so LLM personas produce well under half the response *spread* of the humans they simulate (Figure 2). Evolutionary optimisation moves none of these appreciably.

Under-dispersion is the wrong summary. A single variance ratio hides which variance is missing. Decomposing each answer matrix as $Y_{pj} = \mu + a_p + b_j + (ab)_{pj} + e$ — respondent main effect, item main effect, and the person-by-item interaction that carries individuation — tells a sharper story. The simulator emits nearly the *right total* variance (ratio 0.942 over the 20 cells), but places it in the wrong component: the item main effect takes 81.2% of its variance against 29.8% for humans, while the interaction takes 17.7% against 67.0%. The interaction SD ratio is 0.466, and it is 0 for a constant by construction.

So the simulator is not emitting too little. It is emitting a confident cluster-level *answer key* — which items are endorsable — and almost none of which *person* endorses which item. We recommend reporting the interaction ratio rather than the plain variance ratio, since the two rank systems differently and only the former measures the quantity a personality simulator exists to reproduce.

8.1 What is C2ST actually reading?

A metric that separates simulators from humans at AUC 0.994 invites the objection that the classifier detects formatting rather than personality. A feature ladder (App. D) shows the objection has real force: five content-free response-style scalars

Metric	LLM	Human ceiling	Const. floor
C2ST-AUC ↓	0.994	0.564	0.926
DF ↑	0.783	0.939	0.795
VR (1.0 ideal)	0.431	1.018	0.000
$S_{\text{answer}} \uparrow$ (old)	0.739	0.721	0.796 †

Table 4: The suite over 20 cluster $\times$ model cells, each metric against its human ceiling and constant floor. †: under the old metric the constant floor *exceeds* the human ceiling — the failure this paper documents.

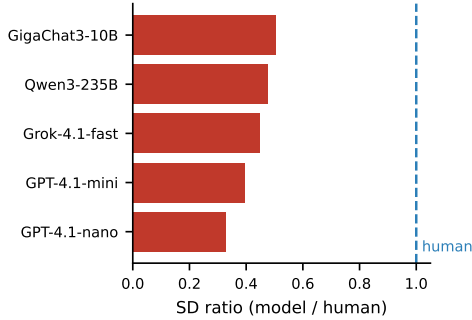


Figure 2: Response *spread* relative to the humans being simulated, as a ratio of standard deviations. Every model is substantially under-dispersed.

already recover most of the separability above chance. It is not only style — ipsatising each respondent, or using facet scores alone, still separates nearly as well — so style and content are redundant routes to the same conclusion. We therefore recommend reporting C2ST with its style-only rung alongside it, so readers can see how much of any separability is carried by response style. A simulator that closed only the style gap would look far better on the headline number while remaining just as distinguishable on content.

The anchors are estimated at $n=40$ per side. Appendix G sweeps n and shows the C2ST human ceiling is entirely finite-sample classifier bias (extrapolating to 0.500) while the DF ceiling is entirely plug-in bias (extrapolating to 1.000, i.e. pessimistic). Below $n \approx 40$ our classifier configuration stops separating at all, so we report every anchor with the n and the settings it was measured at rather than proposing a universal floor.

9 The proposition, reproduced in one experiment

Fitting on half of each cluster and scoring on the other half, two oracles differ only in what they emit: a **mean** decoder (ridge on the ordinal code), which

predicts the conditional mean, and a **residual-resampled** one, the same fit plus a resampled residual, which restores marginal spread. The mean decoder scores $S = 0.802$; the residual-resampled one, which is closer to the response *distribution* on that margin, scores $S = 0.730$ — worse by 7.2 points, against the 7.75 that the GMD — MAD gap predicts. Restoring marginal spread is penalised in this construction, and by very nearly the predicted amount. This construction does *not* settle whether the prompt limits the simulator: it reads a richer code than the ten scores the belief prompts carry, and restoring marginal spread by resampling residuals is not evidence of faithful conditioning. App. F takes that question up on the actual channel.

For scale, this fitted mean decoder beats the constant (0.792) by 0.9 points. That is what one fitted reader achieved here, not a bound on what any reader could achieve.

10 Does this generalise beyond one questionnaire?

Every result so far comes from IPIP-NEO-120, so the objection is that the audit describes that questionnaire rather than the task. We re-test the two *structural* findings — the ones needing no LLM — on an independent instrument, construct and population: the Short Dark Triad (SD3), 17,740 respondents on 27 items.

Both replicate. $GMD \geq MAD$ holds on 27/27 items, with a structural penalty of 8.67 points against IPIP’s 7.75; and the per-item constant again outscores a real respondent from the same population (0.744 vs 0.664) in 200 of 200 bootstraps, matching the inversion of §7. The defect belongs to the metric applied to ordinal questionnaire data, not to the Big Five or to one corpus.

11 Where the variance is lost

Knowing the simulator is mis-dispersed does not say *where* the dispersion goes. We read the model’s own belief before any sampling occurs, on open-weight models served locally.

Two stages. For each persona and item we take the model’s next-token distribution over the answer tokens $\{1, \dots, 5\}$ directly from the logprobs, then sample each item from its own belief and score the resulting population. One implementation detail matters: hybrid reasoning models emit a `<think>` token first — with probability ≈ 1 — so a naive

readout sees no mass on the scale at all; we disable thinking and prefill the assistant turn.

Result. What is collapsed is the *individuating* part of the belief, not the belief’s total spread. In the 2024–25 models both are small; in the 2026 reasoning-first models the total spread reaches or exceeds the human marginal while individuation stays as low as ever.

The statistic matters. The quantity comparable to a human *across-respondent* SD is not the average within-persona conditional SD but the SD of the persona-*mixture* distribution, from the law of total variance $\text{Var}(X_j) = \mathbb{E}_i[\text{Var}(X_j | i)] + \text{Var}_i(\mathbb{E}[X_j | i])$, computed exactly from the retained probability vectors. The within-persona term alone would understate a faithful simulator, which can hold sharp per-persona beliefs and still match human variance through differing persona means.

Model	Params	VR _{with}	VR _{betw}	VR _{tot}
Qwen2.5-7B-Instruct	7B	0.367	0.193	0.430
Qwen2.5-32B-Instruct	32B	0.242	0.220	0.343
Qwen2.5-72B-Instruct	72B	0.367	0.295	0.488
Mistral-Small-24B	24B	0.538	0.177	0.579
QwQ-32B (reasoning)	32B	0.729	0.154	0.753
Qwen3-235B-A22B-2507	235B	0.695	0.258	0.758
Qwen3.5-9B	9B	1.058	0.167	1.078
GLM-4.7-Flash (MoE)	29B	1.209	0.265	1.242
gpt-oss-20b (MoE)	21B	1.121	0.126	1.131
Granite-4.1-8B	8B	0.386	0.207	0.457
Apertus-8B-Instruct	8B	0.927	0.130	0.939
human	—	—	—	1.000

Essentially all probability mass lands on the five valid answer tokens (0.97–1.00), so no model has difficulty *emitting* a number. Belief entropy nonetheless spans 0.42–1.38 nats (maximum $\ln 5 = 1.609$), and we can offer no clean account of it: it tracks neither release date (Granite-4.1, 0.49) nor reasoning-tuning (QwQ 0.97; the standard-instruct Apertus 1.24). The 235B model is the one the audited system used, so this measures that system, not a proxy.

Total dispersion and individuation are decoupled. The decomposition says something the total does not, and the evidence for this is existential rather than correlational. gpt-oss-20b and Qwen2.5-72B invert cleanly: the former has $2.3\times$ the total dispersion of the latter (1.131 vs 0.488) and *less than half* its individuation (0.126 vs 0.295). The rank correlation over the eleven models is $\rho = -0.30$, but its bootstrap 95% interval is $[-0.92, 0.42]$, so we rest nothing on it.

QwQ-32B sharpens it: its dispersion is almost all per-item uncertainty (0.729 of 0.753), and it

individuates less (0.154) than far smaller models.

The Qwen2.5 family isolates this at fixed family and varying scale: individuation rises monotonically, $0.193 \rightarrow 0.220 \rightarrow 0.295$ at 7B, 32B, 72B, while the total does not, moving $0.430 \rightarrow 0.343 \rightarrow 0.488$.

One bound survives across the whole set — a $34\times$ parameter range, six families, three generations, dense and mixture-of-experts, and both standard and reasoning-tuned models: no model exceeds $\text{VR}_{\text{between}} \approx 0.30$. Individuation sits between an eighth and under a third of the human value throughout, and neither the largest model nor the most dispersed one is the best at it.

The persona helps, and still loses. Two questions have run together: does the simulator reproduce the cluster’s distribution, and does it identify *this* respondent? Scoring each persona against that person’s own answers under normalised CRPS, and permuting belief rows among respondents within a cluster — which leaves every population statistic identical while destroying identity — separates them. All eleven models score better on the correct assignment than the permuted one, by 0.004–0.017, every paired interval excluding zero: the persona is used, in the right direction. All eleven nevertheless remain worse than a cluster empirical prior fitted on *other* respondents — a comparator that uses cluster membership but no persona — by 0.036–0.095.

The two facts reconcile exactly, and without any fitted reference. Writing I for the identity gain above, $\bar{q}$ for the model’s within-cluster mean CDF and D_Q for the across-persona variance of its CDFs,

$$R(P_0) - R(Q) = I - D_Q - [R(\bar{q}) - R(P_0)], \quad (3)$$

an identity in measured quantities, exact on our data to machine precision. It locates the loss. The bracket — how much worse the model’s *mixture* is than the prior — is 0.038–0.090, whereas the net identity contribution $D_Q - I$ lies within ± 0.005 for every model, below a tenth of that model’s mixture excess. Spreading predictions across personas can help or hurt slightly; what decides the comparison is that the mixture itself is wrong.

The prompt is not what caps individuation. A persona is individuated only through a five-word quantisation of ten numbers, but that channel carries more than the models extract: a supervised conditional-mean decoder fitted on held-out respon-

dents reaches 0.482 on these same personas, where the best model delivers 0.295, and no two personas share a code (App. F). The decoder is a reference, not a bound — a mis-calibrated simulator could exceed it by dispersing wrongly — so it shows the conditioning carries signal.

How much of it points the right way. VR_{between} measures how far apart a simulator places its persona means, not whether that spread identifies the right respondent. Correlating each model’s persona means against the oracle’s over the forty personas — both on the corpus’s recoded scale — gives $r = 0.16\text{--}0.52$. The aligned amplitude $A = \frac{1}{J} \sum_j r_j \text{sd}(M_j)/\text{sd}(H_j)$, in units of human item SD, is $0.033\text{--}0.183$, against 0.498 $[0.461, 0.517]$ for real respondents: the best reaches about a third of that, the weakest under a tenth. It separates models VR_{between} ranks together — GLM-4.7 and Qwen3-235B sit 0.007 apart in spread, yet differ in A by $+0.017$ $[+0.009, +0.025]$ — and rises with scale across Qwen2.5 in three non-overlapping steps ($0.067 \rightarrow 0.116 \rightarrow 0.183$).

A simulator can therefore look more dispersed simply by being less confident: VR_{total} confounds uncertainty with individuation, and VR_{between} is what a persona claim should be judged on.

One limit, plainly: the strong form of the claim below — that no sampling-side remedy could help — needs total dispersion well under the human marginal, which is false for the largest 2024–25 model (0.758) and for the three that *exceed* it (1.078, 1.131, 1.242); there, spread is present and is lost afterwards. What stays limited at every scale is *individuation*.

Two consequences follow. For the low-dispersion models the spread is not discarded at decoding; it was never present, so recovering it there means *adding* dispersion, not revealing it: $p^{1/T}$ lifts Qwen2.5-7B from 0.43 to 1.02 at $T=5$ while supplying none of the alignment A measures. And since all mass lands on valid answer tokens, accounts blaming mode collapse on difficulty *emitting* numbers are unsupported here.

12 Is $k=4$ a partition at all?

The systems we audit condition on a $k=4$ k -means partition of the 30 facet scores. We re-derive it and test $k \in \{2, \dots, 15\}$ under three preprocessings. Our $k=4$ solution reproduces the published labels ($\text{ARI} = 0.946$), confirming the pipeline.

No criterion supports $k=4$: silhouette is 0.082

there and falls with k , and Davies–Bouldin and the gap statistic select no interior k . Bootstrap stability looks like a defence ($\text{ARI } 0.903\text{--}0.977$ for $k \leq 5$) until structureless surrogates matched to the facet covariance reproduce it exactly (0.911 and 0.921 against the real data’s 0.910), while a separated control confirms the test has power. At $n=410,168$ bootstrap agreement measures estimator variance, not multimodality (App. E). Since stability-based selection is common, the general point is worth stating: at large n , bootstrap agreement must be read against a covariance-matched null or it certifies nothing.

13 Related work

Personality conditioning of LLMs is commonly validated by administering psychometric inventories to the conditioned model (Serapio-García et al., 2023; Jiang et al., 2024), and prompt sensitivity of the resulting traits is well documented (Dong et al., 2025; Gupta et al., 2024). Individual-level simulation from interview transcripts achieves high behavioural fidelity (Park et al., 2024) at a per-person cost that motivates the cluster-level shortcut we study. Evolutionary and search-based prompt optimisation (Guo et al., 2024; Prasad et al., 2023; Zhou et al., 2022) supplies the optimiser. Our contribution is orthogonal to all of these: we show the objective they are scored against is maximised by a degenerate solution.

Classifier two-sample tests are standard for comparing distributions (Lopez-Paz and Oquab, 2017), and the failure mode we identify — an absolute-error objective minimised by a conditional central tendency — is why such losses blur predictions in generative modelling.

14 Conclusion

Cluster-level LLM personality simulation is evaluated with a metric that a constant vector wins and a real human loses. We proved why, measured the structural penalty at 7.75 points, confirmed the inversion in 20 of 20 matched cells, and proposed a suite that orders humans above constants. Beneath it the simulation is thinner still: individuation that identifies the right respondent reaches about 37% $[34, 40]$ of the human reference. Correct persona assignment helps every model and none of them enough: an exact decomposition puts the loss in the mixture, not in identity. Score with $S_{1/2}$; report floor and ceiling beside every number.

Limitations

Our audit covers one corpus (IPIP-NEO-120), one clustering pipeline, one optimisation family, and five models; the belief-level measurement of §11 covers eleven open-weight models spanning 7B–235B, six families and three generations, and the cross-instrument replication of §10 covers only the two structural findings, since SD3 carries no simulator runs; since it requires logprob access the API models in the audit do not provide; the within-family trend rests on three points and the cross-family comparison on eleven, so both are directional rather than fitted laws; all models are read with thinking disabled and the assistant turn pre-filled, and the four least confident of them (belief entropy above 1 nat) should not have their above-human total dispersion read as calibrated uncertainty, though low confidence cannot inflate VR_{between} , which is computed from persona means; two elicitation details proved to matter more than we expected — reading these models through a chat-completions server returned almost no mass on the answer tokens where the forward pass recovers 0.99–1.00, and harmony-format models require the assistant turn to open a channel before the pre-fill, without which mass on the scale falls to 0.39 — so a null result in this literature may report a read-out rather than a model; precision varies across the models we report — nine run at bf16, gpt-oss-20b ships MXFP4 and is dequantised to bf16 at load, and Qwen3-235B is an FP8 checkpoint served at FP8 — and a further FP8 checkpoint resisted measurement through both available routes, so we make no claim that individuation is invariant to quantisation; the proposition is general to the absolute-error metric in (1), but the empirical magnitudes are specific to this setting. The model answers we re-score were produced by prior runs we did not control, at temperatures we did not set, and one released run was committed with entirely missing answers and is excluded. Our human ceiling is a single same-cluster respondent, which underestimates what a well-specified individual-level twin could achieve; we make no claim about individual-level methods. C2ST-AUC depends on classifier capacity and on $n=40$ per side, so its absolute value should be read as a lower bound on separability, and §8.1 shows most of what it reads is response style; we fold the AUC as $\max(\text{AUC}, 1 - \text{AUC})$, which places the null above 0.5 (0.517 signed against 0.569 folded) and makes our small- n floor a property of this clas-

sifier rather than of C2ST; and DF treats items independently and so ignores within-respondent answer dependence. The corpus is self-reported, skewed toward respondents aged 18–25 and toward the USA, UK and Canada, so neither the clusters nor the ceilings should be read as representative of any broader population. Finally, we show current personas fail this evaluation; we do not show that any method can pass it.

Ethics statement

We use a public, de-identified corpus of self-reported personality questionnaire responses and add no new human-subjects data collection. Our results are cautionary: they indicate that current cluster-level personality simulations are readily distinguishable from real respondents and under-represent human response variance, and so should not be used as substitutes for human participants in research, product, or policy decisions. Reporting an evaluation number without its constant floor risks overstating the fidelity of synthetic respondents, which is the specific harm this paper aims to reduce. Simulating demographic or personality groups also risks reinforcing stereotypes when cluster descriptions are treated as characterisations of real people rather than as statistical summaries.

A Reproducibility

Every analysis in this paper is deterministic given the released corpus and the committed run artefacts, and requires no LLM calls except the belief-level measurement of §11. We ran the full analysis pipeline independently on two clusters with different operating systems, Python versions and library versions (scikit-learn 1.2 vs 1.7, pandas 2.1 vs 3.0). Every reported figure we compared — the λ^* values, the orientation deltas, the variance components, and the null-calibration ARIs of App. E — agreed to the digits we print, except the $k=4$ reproduction ARI, which is 0.946 on one environment and 0.951 on the other. Code and per-cell results are released, including the full belief simplex for every model and encoding variant we report: (40 personas $\times$ 120 items $\times$ 5) per cluster, with the persona ordering fixed by the released corpus so every reported number can be recomputed from the artifacts alone. Runs that failed the validity guard ship too, marked invalid, so the record shows what was measured and rejected. One transform maps model answers onto the corpus’s recoded scale and

is imported by every analysis that compares the two.

B Split sensitivity of the baselines

Re-drawing which 100 respondents are sampled at a fixed 60/40 ratio moves the parameter-free cluster-mean baseline by 0.54–0.77 points within a cluster over 20 draws, and the four-cluster average by 0.23 (pooling clusters gives 1.33, mixing cluster differences into split variation). Prior work’s gain is +0.72 points on one fixed split. This is context for how far a split can move a score, not a test of that gain: variation in baseline *levels* does not bound a *paired* improvement.

C What is a persona worth, in respondents?

“It loses to a constant” invites the reply that the constant is a strawman. So we restate the finding in a unit that is harder to dismiss: how many real respondents you would have to sample before a plug-in estimate of the cluster’s response distribution matches what the persona gives you.

Drawing m real respondents, forming the per-item empirical distribution, and sampling a synthetic population from it yields distributional fidelity 0.723 at $m=1$, 0.797 at $m=2$, 0.915 at $m=21$ and 0.923 at $m=34$. The curve is evaluated at the same sizes as the audited scores — forty synthetic against forty held-out humans — because DF depends on sample size, so a curve built at any other size cannot calibrate them. Reading the audited systems off it:

System	DF	Resp.-equiv.
Evolutionary LLM persona	0.783	2
Constant cluster mean	0.795	2
Human ceiling	0.939	off-curve

An evolutionary-optimised persona is worth about **two** real respondents: two is the smallest evaluated m whose mean DF exceeds the persona’s 0.783, and the constant sits at the same place. This is a threshold on the mean curve, not an equivalence test. The human ceiling of 0.939 is not crossed anywhere on the evaluated grid, up to $m=500$ (0.938), so we do not convert it into respondent units — a statement about the grid we searched, not a proof that no m reaches it. This is the same result as §4 in different units, and it is the version we would put to a practitioner weighing synthetic respondents against a small pilot survey.

D The C2ST feature ladder

A metric that separates simulators from humans at AUC 0.994 invites an obvious objection: perhaps the classifier is detecting formatting rather than anything about personality. We test our own metric with a feature ladder, each rung scored identically against the same anchors. Orientation is handled per feature family: style features describe how a respondent uses the scale, so both sides sit on the literal scale the item was asked on, while content features compare trait-oriented quantities. The full-item rung reproduces the suite’s 0.994.

Features	LLM	Human	Const.
Full 120 items	0.994	0.564	0.926
5 style scalars only	0.894	0.584	0.988
Items, ipsatised	0.989	0.566	0.986
30 facet scores only	0.991	0.565	0.908
Forward-keyed only	0.990	0.568	0.810
Reverse-keyed only	0.993	0.583	0.843

Five content-free scalars — per-respondent mean and standard deviation, extreme -response rate, longest identical run, and forward-versus-reverse consistency — attain 80% of the full above-chance AUC on their own. The objection has real force, and we state it plainly: **response style alone reproduces most of the separability**, which is a statement about what suffices, not a decomposition of what the full classifier uses.

It is not only style. Ipsatising each respondent (removing their own mean and standard deviation, i.e. the two largest style features) still leaves AUC 0.989, and facet scores alone give 0.991. Style and content are redundant routes to the same conclusion rather than one masking the other. The human ceiling is also stable at 0.56–0.58 across every rung, so the anchor does not move with the feature set.

E Cluster validity in full

No *separation* criterion supports $k=4$. Silhouette falls overall from 0.134 at $k=2$ to 0.053 at $k=15$, with minor upward steps, taking 0.082 at $k=4$; Davies–Bouldin is also minimised at $k=2$; the gap statistic increases monotonically and selects no interior k . A silhouette below 0.1 indicates essentially no separated structure.

Stability looks at first like a defence: bootstrap ARI over 40 resamples is 0.977, 0.952, 0.932, 0.903 at $k=2, 3, 4, 5$ and falls away after (0.797, 0.581), so $k=4$ looks close to the largest reproducible k .

That reading does not survive a null calibration. Rebuilding the pipeline on structureless surrogates

— a Gaussian matched to the real 30×30 facet covariance, and a copula matched to the covariance and every per-facet marginal — reproduces the stability number exactly: ARI 0.911 and 0.921 at silhouette 0.075 and 0.078, against the real data’s 0.910 at 0.082. A separated 4-component control confirms the test has power (silhouette 0.672, ARI 1.000).

At $n=410,168$ any smooth unimodal cloud with this covariance is partitioned reproducibly, so ARI ≈ 0.91 measures estimator variance, not multimodality, and neither criterion supports $k=4$. The clusters are not demographic artefacts either (Cramér’s V of 0.055–0.057 against sex, age band and country); they are a reproducible discretisation of a continuum, which bounds the cluster-conditional signal any persona can exploit. Since stability-based model selection is common, the general point is worth stating: at large n , bootstrap agreement must be read against a covariance-matched null or it certifies nothing.

F What the persona channel carries

A persona reaches the model only through `get_modifier_bisect`, which writes each score as one of five words on width-20 bins; each cluster’s genotype names five traits and five of the thirty facets, and every other sentence of the system prompt is shared. Ten words therefore carry the respondent’s identity, and two respondents sharing those ten bins would receive byte-identical prompts and so must receive identical beliefs.

We estimate what that channel permits. Within each cluster we fit $\mathbb{E}[X_j \mid \text{code}]$ on respondents *excluding* the forty personas, then evaluate on those forty, so the quantity is directly comparable to $\text{VR}_{\text{between}}$ and is never fitted on its own evaluation set.

Conditioning available to the model	Dec. SD
Ten scores, five words (<i>audited system</i>)	0.482
same, with boosted trees	0.483
Ten scores, written as numbers	0.512
All 35 scores, five words each	0.707
All 35 scores, written as numbers	0.726
<i>best model observed</i>	0.295

Three things follow. The quantisation is not the binding constraint: writing numbers instead of words buys 0.03. The channel is not degenerate: all forty personas in all four clusters receive distinct codes, so no model is forced to answer two personas identically. And a boosted-tree decoder, which can use interactions, lands in the same place

(0.483 against 0.482); that is consistent with the additive fit being close to the conditional mean, though two similar SD ratios do not prove it.

The oracle is fitted on this population and so knows its code-to-response mapping, which a zero-shot model does not; we therefore do not claim that a model *ought* to reach 0.482. The comparison rules out one explanation only — that the conditioning carries too little information to individuate — and that is the explanation the flatness of $\text{VR}_{\text{between}}$ across eleven models most invites.

Uncertainty on A. Persona rows are resampled jointly across the model, the decoder and the actual respondents, so paired comparisons stay paired; the decoder fit is held fixed, so the intervals describe persona-sampling uncertainty rather than fit uncertainty. Over 2000 draws the human reference is 0.498 [0.461, 0.517] and the eleven models span 0.033 [0.028, 0.036] for Apertus-8B to 0.183 [0.168, 0.193] for Qwen2.5-72B. The intervals are informative enough to separate models that $\text{VR}_{\text{between}}$ ranks together: Qwen2.5-32B and Granite-4.1 differ by 0.013 in spread but their amplitudes do not overlap, [0.105, 0.122] against [0.071, 0.089]. Comparisons are made on the paired draws, not by asking whether marginal intervals overlap — two overlapping marginals can still have a difference interval excluding zero, as GLM-4.7 and Qwen3-235B do (+0.017 [+0.009, +0.025]) — and Qwen2.5-32B exceeds Granite-4.1 by +0.034 [+0.028, +0.040] despite a 0.013 gap in spread.

Where the shortfall falls. Each cluster names five facets, so twenty items have their own facet stated and one hundred are covered only at trait level. The named items carry $1.51\times$ more predictable-from-code signal (an SD ratio) (reference 0.671 against 0.444), and 10 of 11 models individuate them more in absolute terms, Qwen3-235B being the exception (0.224 named against 0.264 elsewhere). Measured against those references the ratio is *lower* where the prompt is most specific — 0.353 on the named items against 0.433 on the rest, in the same direction for 10 of 11 models — but we report this descriptively and draw no causal conclusion from it, for two reasons. A facet score is computed from its own four items, so $\mathbb{E}[X_j \mid \text{facet}]$ is close to an arithmetic identity that a model without the scoring key cannot invert, while traits have the same structure diluted over twenty-four items; the named-item reference is in-

flated by more than the trait-level one, which alone could produce the gap. And the eleven models share personas, references and prompt structure, several within one family, so they are not independent replicates and we do not attach a p -value to the count.

Widening the channel does not fix it. The obvious remedy is to send more: we re-ran three models with every other word of the prompt held fixed and the ten modifier phrases replaced by the numbers themselves. The effect is inconsistent in magnitude — VR_{between} rises for Granite-4.1 ($0.207 \rightarrow 0.246$) and gpt-oss ($0.126 \rightarrow 0.146$) but falls for Apertus ($0.130 \rightarrow 0.096$) — and, using the same joint draws so the comparison is paired, every model moves resolvably: A shifts by $+0.053$ [$+0.045, +0.059$] for Granite-4.1 and $+0.008$ [$+0.005, +0.012$] for gpt-oss but -0.013 [$-0.016, -0.009$] for Apertus. The effect is therefore not noise; it is real and model-dependent in sign. It is also inconsistent in direction, the aligned amplitude moving $0.081 \rightarrow 0.134$ for Granite-4.1 and $0.039 \rightarrow 0.047$ for gpt-oss but $0.033 \rightarrow 0.020$ for Apertus. Granite gains on both counts — its correlation with the oracle rises from 0.285 to 0.453 — so for that model the richer encoding is a real improvement, while Apertus loses spread and direction together. On three models a richer encoding is therefore a plausible but unreliable remedy, and we report it as a pilot rather than a recommendation.

The alignment does not depend on the oracle. Because r is measured against a fitted target, an oracle missing structure the models track would understate it. It does not. Boosted trees on the same code change r by at most 0.012 and a ridge on the continuous scores by at most 0.034; an oracle given all thirty-five scores lowers r substantially, by up to 0.168, which is the expected direction since models only ever see the ten-score code. The ordering of models is unchanged throughout. Two independent computations agree where theory demands it: the ceiling as a standard-deviation ratio is 0.482 and the oracle’s correlation with actual respondents is 0.491, which coincide for a conditional mean. The check that assumes no oracle at all — correlating each model’s persona means directly against the forty respondents’ own answers — gives 0.076–0.248 against the oracle’s 0.491, so models recover between a sixth and a half of the attainable person-level alignment.

G Finite-sample behaviour of the anchors

Both anchors are estimated from $n=40$ per side, which for a classifier over 120 features is a small-sample regime. We sweep n on human-only data, where the true answers are known: two disjoint samples from the same cluster must be indistinguishable, so the honest C2ST value is exactly 0.5 and any excess is estimator bias.

n	20	40	80	160	320	640
C2ST human	—	0.574	0.542	0.536	0.527	0.517
DF human	0.915	0.939	0.957	0.969	0.978	0.985
C2ST const.	—	0.914	1.000	1.000	1.000	1.000

Two corrections follow, in opposite directions, and both land on their theoretical values. Fitting $y = y_{\infty} + c n^{-1/2}$ over the non-degenerate range gives an extrapolated C2ST human ceiling of **0.500** and an extrapolated DF human ceiling of **1.000** — exactly what two samples from the same distribution must score. So the C2ST ceiling of 0.574 at $n=40$ is *entirely* finite-sample classifier bias and should be read with its n attached, while the DF ceiling of 0.939 is *entirely* plug-in Wasserstein bias and is pessimistic. Correcting the latter widens the gap to the simulator’s 0.783 rather than narrowing it.

Below $n \approx 40$ *this* classifier stops separating even the constant from real humans (the dashed entries above are the degenerate 0.5). That floor is a property of the configuration we used, not of C2ST: relaxing the minimum leaf size from 20 to 5 restores separation at $n=20$. We therefore recommend reporting every anchor together with the n and the classifier settings it was measured at, and treating any small- n floor as something to establish per configuration rather than to inherit from us.

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